

FIRST. / LAST.

Make row order part of the request

01 / THE SAS EXAMPLE

```
data test;
  input id group $ value;
  datalines;
1 A 10
2 A 20
3 B 30
4 B 40
5 B 50
6 C 60
;
run;
data flagged;
  set test;
  by group;
  first_flag = first.group;
  last_flag = last.group;
run;
proc print data=flagged noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

State the ordering assumptions that define a row-based result.

These six records are sorted by group. FIRST.group and LAST.group mark each group's beginning and end. Group C has one member, so its row receives both flags.

EXPECTED RESULT

id	group	value	first_flag	last_flag
1	A	10	1	0
2	A	20	0	1
3	B	30	1	0
4	B	40	0	0
5	B	50	0	1
6	C	60	1	1

Scope: one nonmissing grouping variable and the supplied sorted rows. Both flags are 1 for a one-row group. All six rows remain in the output.

Reported outputs: A 3/3 | B 3/3. Compliance: see p. 3.

Reference checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.
Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

All A runs chose group positions

Run	Method	Output match
A1	groupby().cumcount()	Yes
A2	groupby().cumcount()	Yes
A3	groupby().cumcount()	Yes

All three report correct flags; none included assertions.
Position 0 marks first; position size - 1 marks last.
Agreement in three reports does not guarantee future runs.

SOURCES: results-20260921.csv; trial-log-20260921.md; evidence-review-20260922.md

WHY CHOICES REMAIN OPEN

- 1

The source supplies the order

The BY statement and sorted rows support a correct translation even with a short request.
- 2

More than one method works here

Group positions and adjacent-row comparisons produce the same flags for these sorted, nonmissing groups.
- 3

Correct flags still need checking

Keep all six rows in order. Test both lists, including interior rows and the one-row group C.

ANTI-PATTERN: "KEEP FIRST AND LAST"

That asks for filtering. This program adds flags to every row. Filtering group boundaries would discard row 4 and change the task.

Verify six ordered rows, not just group boundaries.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
Show code in this response. Do not create or run files, or read
existing Python files.

- Recreate id, group, and value. Keep all six rows in their order.
- These rows are already sorted by group, with no missing groups.
- Add first_flag and last_flag as integer 0/1 columns.
- first_flag is 1 on the first row or when group differs from
  the previous row. Compare with group.shift(1).
- last_flag is 1 on the last row or when group differs from
  the next row. Compare with group.shift(-1).
- Handle the first and last rows explicitly. Do not filter rows
  or reorder records within groups.
- Include assertions for id = [1, 2, 3, 4, 5, 6],
  first_flag = [1, 0, 1, 0, 0, 1], and
  last_flag = [0, 1, 0, 0, 1, 1].
- Return one complete script. Compute the flags from group;
  do not hard-code them. Explain why row 6 has both flags.
```

WHY THESE DETAILS HELP

- 1

Preserve rows and their sequence
One sorted, nonmissing group key; keep every row. Group C must receive both flags.
- 2

Make endpoint handling explicit
B1 includes endpoint checks. B2/B3 rely on missing comparisons, omitting this prompt requirement.
- 3

Separate output from compliance
All B outputs and assertions are reported as passing. Only B1 shows explicit endpoint handling.

REPORTED B RUNS / REVIEWED EXCERPTS

Run	Output / assertions	Explicit endpoints
B1	Yes / passed	Yes
B2	Yes / passed	No
B3	Yes / passed	No

CHECK IT YOURSELF
Check both lists, row order and explicit endpoints.
Nullable strings can propagate NA at shifted endpoints.

One behavior, three implementations

Flag the beginning and end of each sorted group, keeping every row. Endpoints are explicit, including a data set with one row.

SAS / SOURCE BEHAVIOR

```
%macro transform(in=, out=);
data &out;
  set &in;
  by group;
  first_flag = first.group;
  last_flag = last.group;
run;
%mend;
```

Macro transform supplies source/result names. BY group creates FIRST.group and LAST.group, including both endpoints of a singleton.

PYTHON / PANDAS

```
out = df.copy(deep=True)
group = out['group']
first = group.ne(group.shift(1))
last = group.ne(group.shift(-1))
if len(out):
    first.iloc[0] = True
    last.iloc[-1] = True
out['first_flag'] = first.astype(int)
out['last_flag'] = last.astype(int)
```

translate checks sorted, nonmissing group values before these comparisons. Explicit iloc endpoints also work with nullable strings.

R / BASE

```
g <- out$group
n <- length(g)
first <- last <- integer(n)
if (n > 0L) {
  first[1L] <- 1L
  last[n] <- 1L
  if (n > 1L) {
    change <- g[-1L] != g[-n]
    first[-1L] <- as.integer(change)
    last[-n] <- as.integer(change)
  }
}
out$first_flag <- first
out$last_flag <- last
```

out <- df precedes this excerpt. translate validates sorted, nonmissing groups. Integer vectors and explicit endpoints handle a singleton.

Output and instructions both matter
Historical B2/B3 summaries match the fixture but omit explicit endpoint handling. This reference honors that requirement; trial scores stay unchanged.

Keep the whole row sequence
These flags annotate every record. Filtering to only boundary rows drops id 4 in the original fixture and changes the operation.

SAS BY-group processing

pandas missing values

R integer vectors

Three checks that earn the pass

Use the same inputs and expected behavior in each language. Check the contract as well as the values.

T1 / Original known-answer fixture

INPUT / CONDITION

Groups A,A,B,B,B,C in original id order 1-6.

EXPECTED

first=[1,0,1,0,0,1]; last=[0,1,0,0,1,1].

WHAT THIS CATCHES

Wrong boundary direction, lost interior rows or mishandled one-row group C.

T2 / Added one-row data set

INPUT / CONDITION

One row: id=41, group=Z, value=99; Python uses nullable strings.

EXPECTED

Both integer flags equal 1; exactly one row remains.

WHAT THIS CATCHES

Reliance on NA comparisons, endpoint omissions, and two-row minimum assumptions.

T3 / Preserve rows and integer flags

INPUT / CONDITION

Original fixture with non-default Python/R row labels.

EXPECTED

Input and source order unchanged; complete correct integer 0/1 flags.

WHAT THIS CATCHES

Index-label endpoints, row filtering/reordering, input mutation or string/boolean flag outputs.

REFERENCE CODE + TESTS: [module-09-firstvar/appendix/](#)

Run: `python tools/run_appendix_tests.py --modules 09`

New reference code, separate from historical trials. SAS checks await execution; scope is stated in the companion README.